



COPILOTTRAINING
TECHNICAL BRIEFING

Agentic Economics: The Business Case

| The ROI Framework for Agentic AI Adoption

What it takes to turn the \$2/hour opportunity into P&L reality

CopilotTraining · Executive Talk



A Three-Part Executive Briefing Series



The Labor Multiplier

What agents can actually do — the 67% of delivery labor outside the code editor that AI is built to handle at a fraction of human cost



Agentic Economics

How to calculate and capture ROI — agent compute costs, infrastructure investment requirements, and the honest payback math



Agentic Delivery

The operating model for safe scale — instrument infrastructure, pilot oversight, and the governance that enables the throughput multiplier



Each briefing is complete on its own. Together they form a full strategic framework for agentic AI investment.

Three Briefings, Three Decisions

The Labor Multiplier

Which of the 67% of delivery labor is ready to automate — and what organizational investment does capturing it require?

Opportunity Decision

Agentic Economics

What is the realistic ROI timeline, and what infrastructure makes the \$2–5/hour agent arbitrage accessible at scale?

Capital Decision

Agentic Delivery

What governance infrastructure — instruments, oversight model, no-fly zones — must exist before agent programs can operate safely?

Operating Model Decision

These are the three decisions the briefing series is designed to drive. This talk addresses one of them.

The Complete Decision Framework



Identify the Opportunity

The Labor Multiplier: 67% of delivery labor has no AI coverage — discovery, compliance, documentation, and validation are ready for agents



Quantify the Investment

Agentic Economics: agent compute at \$2–5/hour creates a structural cost arbitrage — the ROI depends on the verification infrastructure



Enable Safe Operation

Agentic Delivery: the instrument panel, pilot model, and governance structure that make the multiplier achievable at enterprise scale



The Labor Multiplier identifies what is possible. Agentic Economics quantifies the return. Agentic Delivery specifies the infrastructure that makes both achievable.

PART 1

The Arbitrage

Cost arbitrage and the work spectrum

Cost gap and the three categories of work it reframes



PART 2

The Capture Problem

Infrastructure gaps block P&L capture

Five barriers, one root cause



PART 3

Quick Wins

Issue lifecycle as the proof point

One workflow, 3.6-day payback, self-funding

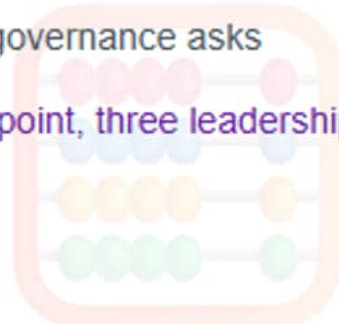


PART 4

The Calculation

Investment scenario and governance asks

J-curve, dashboard checkpoint, three leadership asks



The Labor Arbitrage Opportunity

Agent compute at \$2–5/hour vs \$130/hour for senior engineers creates a structural cost differential.



The Cost Gap

\$5/hr vs \$130/hr — a 20–65x fully-loaded differential



Three Work Buckets

High-value, automatable, toil — each priced differently



The DORA Signal

90% AI adoption, 24% output trust — the gap follows

Agent compute cost vs senior engineer (fully loaded)

↳ \$2–5/hr vs \$100–130/hr

The Cost Differential That Reframes Engineering Economics

Fully-loaded agent compute vs senior engineer labor, 2026

20–
65x

fully-loaded labor cost differential — agent compute
vs senior engineer

Agent: \$2–5/hr (API + orchestration + monitoring) | Senior engineer:
\$100–130/hr (salary, benefits, overhead) — BLS 2024

💡 The question is not whether agents are cheaper. The question is what work can move to \$2–5/hour, and what infrastructure is required to capture it.

🤖 Agent Compute

\$2–5/hr for production-grade agentic workflows including API calls, orchestration, and monitoring

💼 Mid-Level Developer

\$85–110/hr fully loaded — 17–55x more expensive than agent compute for the same task time

🌐 Offshore Developer

\$25–85/hr — still 5–42x more expensive than agent compute even at the lowest offshore rates

🎯 The Strategic Question

Which engineering hours can move to the \$2–5/hr tier — and what infrastructure is required?

Three Categories of Engineering Work — Three Price Points



High-Value Work

Architecture, strategy, and complex debugging — judgment-intensive work at \$100–150/hr.

Customer conversations

Trade-off evaluation

System design decisions



Routine Work

Code review, test writing, documentation — pattern-based work already accessible at \$2–5/hr

Bug investigation

Dependency updates

Compliance checking



Pure Toil

Issue triage, audit prep, status reports — rule-following work best suited for \$2–5/hr agents

Duplicate detection

Format enforcement

Data aggregation



DORA 2025: 90% of organizations have adopted AI tools — only 24% trust AI outputs enough to deploy without extensive review. The capture problem is structural.

The Capture Problem

AI pilots are approved and the P&L is unchanged. Five barriers — one root cause: no observability infrastructure.



McKinsey Validation

5% of pilots reach material P&L improvement



The Perception Gap

Teams believe 20% faster; measured 19% slower



Five Barriers, One Root

No instrumentation means no capture

McKinsey QuantumBlack 2024 – AI pilot success rate
↳ 5% deliver material P&L improvement

Most Organizations Have Been Cautious — Correctly

McKinsey QuantumBlack 2024: P&L impact rate for AI pilots across organizations

5%

of AI pilots deliver material improvement to
organizational P&L

*McKinsey QuantumBlack: The State of AI in 2024 — a half decade
in review*

✓ The Correct Instinct

Caution about AI ROI has been proportionate — most pilots have not delivered what they projected

🔍 A Diagnostic, Not a Verdict

The 95% shortfall identifies where the model breaks down — a blueprint for what to fix

🏗️ Infrastructure Is the Variable

Pilots reaching the 5% share one common factor: automated verification before agents touch code.

💡 The gap between pilots and P&L impact is not model quality. It is measurement infrastructure — teams are flying without instruments.

Flying Blind AND Confident

METR RCT 2025: experienced developers on complex real-world open-source tasks

20% →
-19%

self-reported productivity gain vs independently
measured slowdown on complex tasks

*METR RCT 2025 — experienced open-source developers; fewer
than 44% of AI suggestions accepted on complex repos*

🗨️ The Perception Gap

Developers self-reported 20% faster. Measured: 19% slower on complex, unfamiliar codebases.

🔍 Where the Slowdown Occurs

Iteration overhead explains the gap. Fewer than 44% of suggestions accepted on complex repos.

✅ Where Gains Are Reliable

Well-scoped routine tasks with clear requirements and automated verification show consistent gains.

💡 Organizations without measurement infrastructure cannot distinguish confidence from performance — and cannot course-correct when pilots underdeliver.

Five Barriers With One Root Cause

The most common reasons AI investment does not reach the P&L — all trace to missing infrastructure

Verification

No automated tests means every agent output needs full human review.

Verification Gap

Tacit Context

Undocumented context is invisible to agents — wrong assumptions follow

Context Gap

Review Gates

An agent done in 30 minutes still waits 2 days in the review queue.

Flow Gap

No Guardrails

Every task requires negotiating scope before the agent can start.

Governance Gap

Fragmentation

Context scattered across tools means agents get partial inputs.

Integration Gap

All five barriers trace to one diagnosis: no observability infrastructure. That is a solvable problem.

Governance Infrastructure Pays for Itself

✗ Shadow AI Risk

20% of 2025 data breaches originated from shadow AI usage

Unauthorized models bypass security controls and audit trails

\$670K additional cost

Per incident involving shadow AI vs standard breach (IBM 2025)

💡 AI Security Controls

Sanctioned model registry with access policy enforcement

Automated scanning integrated in CI/CD pipelines

Usage monitoring with anomaly detection and audit trails

✓ Measurable Protection

Security investment reduces both breach probability and magnitude

Governance becomes self-funding at the first prevented incident

\$2.2M

avg savings with AI security controls

\$4.44M

global avg breach cost (IBM 2025)

Quick Wins: The Issue Lifecycle Pattern

One workflow, one payback number: issue lifecycle automation proves ROI before the full platform investment.



3.6-Day Payback

\$1M savings, 3.6-day payback — single workflow



The Wedge Principle

Start where work is frequent and approval-light



Self-Funding ROI

Quick wins generate budget for full transformation

Issue lifecycle automation – 50-person team scenario

↳ \$1,001,000/year savings · 3.6-day payback

A Single Automatable Workflow Pays Back in 3.6 Days

Issue lifecycle: research → planning → execution → review (50-person team scenario)

3.6 days

payback period for full issue lifecycle automation —
50-person team scenario

Internal scenario: 20 issues/week, \$100/hr avg loaded rate, agent cost \$3/hr. See README for full assumptions.

💡 The \$10,200 first-year investment generates \$1,001,000 in savings — the highest-ROI entry point into agentic economics for most engineering organizations.

📊 Current State (manual)

\$1,040,000/year — 20 issues/week × 10 hrs each × \$100/hr loaded labor rate

🤖 With Lifecycle Agents

\$39,000/year total — \$7,800 agent API costs plus \$31,200 human validation time

💰 Annual Savings

\$1,001,000/year — 96% reduction in total workflow cost for the same output volume

🔧 Infrastructure Required

4–6 hours of one-time setup. No repository restructuring or

CI/CD rewrite required

From Manual Lifecycle to Agent-Assisted Execution

Four stages, dramatically compressed

✗ Manual Workflow

- 1 **Issue triage**
30 min @ \$100/hr = \$50/issue
- 2 **Planning**
4 hrs research + spec = \$400/issue
- 3 **Implementation**
6 hrs coding @ \$100/hr = \$600
- 4 **Code review**
60 min @ \$100/hr = \$100



\$1,150/issue fully loaded — 11.5 hours per issue lifecycle

✓ With Lifecycle Agents

- 1 **Automated triage**
5 min agent @ \$3/hr = \$0.25/issue
- 2 **Agent-drafted plan**
30 min + 5 min approval = \$3.50/issue
- 3 **Agent implementation**
90 min @ \$3/hr + 15 min review = \$29.50
- 4 **Automated review**
5 min human validation = \$8.33

✓ **\$41.58/issue — 96% cost reduction per issue**

\$1,040,000/year → \$39,000/year for 20 issues/week

Three Principles That Make the Wedge Work



The Wedge

Start where work is frequent, measurable, and approval-light. Issue triage meets all three criteria.

High volume drives fast proof

Measurable before/after baseline

Approval-light reduces bottlenecks



Self-Funding ROI

Quick wins generate the budget for larger infrastructure. \$1M in savings funds the \$1M buildout.

Phase 1 funds Phase 2

Executive buy-in from real data

Team competency as a byproduct



Governance as Gate

IBM data: organizations with governance before scaling see \$2.2M in breach-cost savings.

Audit trails from day one

Access policy before expansion

Shadow AI risk eliminated early

📖 See The Journey to Agentic SDLC for step-by-step setup instructions with working code examples for each stage of the issue lifecycle.

The Leadership Calculation

Year 1 is negative, Year 2 is modeled positive. Infrastructure investment precedes agent deployment for reliable ROI.



The Modeled J-Curve

Year 1 ~\$425K, modeled 240% 3-year ROI



Dashboard Checkpoint

Directional cohort signal — not realized savings



Three Leadership Asks

Instrumentation, governance, workflow redesign

Modeled 3-year ROI — 50 engineers, \$1M infrastructure investment
↳ Year 1: ~\$425K → Year 2: modeled positive → Year 3: modeled 240% ROI

The J-Curve: Year 1 Is Negative

Modeled — 50-person team, \$100/hr average

Year 1

Investment Phase

~\$1M investment. \$635K savings. Net:
~-\$425K.



Year 2

Returns Phase

\$1.486M run rate. No additional
investment. Net: modeled positive.

**TARGET**

Year 3

Compounding

Net: ~+\$2.5M modeled. 3-year ROI:
modeled 240%.



Internal scenario — 13–15% of AI projects achieve these modeled returns.



Impact Dashboard: A Directional Cohort Checkpoint



Monthly Copilot Cost

Per developer, calculated from AI-credit consumption. Tracks spend as agent-first cohort expands.



Modeled Payroll Share

From a selectable salary input — a model input, not payroll data. Results are directional.



Pull Requests / Developer

Average PRs per developer per month. Throughput signal for early-phase vs agent-first cohorts.



Dashboard results are potential and directional — not realized savings or causal proof. Access requires the Copilot usage metrics policy and View Copilot Metrics permission. Corrected 28-day cohort counts apply to the dashboard only, not the usage API or NDJSON exports.

Three Leadership Asks for Monday



Fund Instrumentation

Tests, security scanning, and quality gates must come before agents are deployed.

80%+ coverage on critical paths

CI/CD quality gates

Security scanning automated



Mandate Governance

IBM data confirms \$2.2M in breach-cost savings for organizations that mandate governance first.

Sanctioned model policy

Access and audit controls

Shadow AI detection in CI



Commit to Workflow Redesign

Agents in manual workflows deliver a fraction of the modeled return. Redesign is the multiplier.

Redesign around agent capabilities

Remove approval bottlenecks

Measure iteration counts, not only speed

The question is whether to mandate measurement infrastructure — not whether to approve an AI pilot. The two decisions have different ROI profiles and different failure modes.

The Compounding Advantage Accrues to Organizations That Start Now

Platform maturity, team competency, and measurement infrastructure all compound over time

18–24
mo

head start in platform maturity and team competency for organizations that have already started

Internal scenario modeling — reflects infrastructure investment timeline and organizational learning curve

Platform Maturity Compounds

Infrastructure scales with headcount. Marginal cost per additional engineer approaches zero.

Team Competency Compounds

Agents improve with richer context. Documentation and test coverage increase agent output quality.

Measurement Infrastructure Scales

The verification pipeline built for 50 engineers works for 500. The investment amortizes at scale.

💡 The mandate is measurement infrastructure. The ROI timeline begins when instrumentation begins — not when the first agent is deployed.

From Labor Arbitrage Theory to P&L Reality

Before

Agent cost advantage identified but no mechanism to capture it

AI pilots approved but P&L impact unmeasured

Developers subjectively faster with no objective baseline

Infrastructure investment deferred as optional cost

After

Verification pipeline creates automated trust for agent output

Issue lifecycle automation delivers \$1M annual savings (scenario)

Measurement surfaces where gains are real and repeatable

Instrumentation investment pays back in 12–18 months (modeled)

20–65×

labor cost differential — agent compute
vs senior engineer

\$1M/yr

savings from issue lifecycle (50-person
team, modeled scenario)

240%

modeled 3-year ROI with proper
infrastructure investment

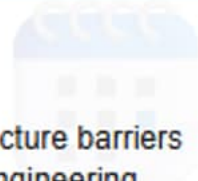
✓ WHAT YOU CAN DO TODAY

Today



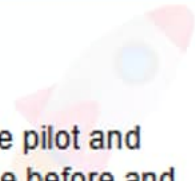
- Read the Copilot impact dashboard docs on cohort access and permission requirements
- Calculate current issue lifecycle cost: $\text{issues/week} \times \text{hours/issue} \times \text{loaded hourly rate}$
- Identify one high-volume, approval-light workflow as the pilot candidate

This Week



- Map the five infrastructure barriers against the current engineering environment
- Request the Copilot usage metrics policy and View Copilot Metrics access
- Scope a 4–6 hour setup for issue triage automation in one repository

This Month



- Run the issue lifecycle pilot and measure time-to-triage before and after
- Build the business case using the internal modeled J-curve (Year 1 ~\$425K)
- Present three leadership asks: fund instrumentation, mandate governance, commit to workflow redesign

🔑 Key Takeaway

Measurement infrastructure is the prerequisite — instrumentation investment precedes agent deployment for reliable ROI.

 Research & Benchmarks

[McKinsey QuantumBlack: State of AI 2024 — 5% material improvement rate](#)

Foundational benchmark for AI pilot P&L success rates

[METR RCT 2025 — developers 19% slower on complex tasks with AI](#)

Perception vs measurement gap on complex real-world codebases

[DORA: ROI of AI-Assisted Development, 2025](#)

Organizational adoption vs output trust gap data

[Yetistiren et al.: AI code correctness — Copilot 46%, arXiv 2023](#)

Benchmark correctness rates for AI code generation tools

[GitHub + Accenture: 84% more successful builds, 2024](#)

Enterprise RCT on structured agentic development workflows

 Related Talks

[The Agentic Labor Multiplier](#)

 Cost & Economics Data

[IBM Cost of a Data Breach, 2025 — \\$4.44M avg, \\$2.2M AI security savings](#)

Governance ROI data; 20% of 2025 breaches from shadow AI

[OpenAI API Pricing, 2025 — basis for \\$2–5/hr agent cost range](#)

Agent compute cost foundation for the labor arbitrage model

[BLS Occupational Employment: Software Developers, 2024](#)

Baseline for fully-loaded engineer labor cost ranges

[GitHub Changelog: Copilot impact dashboard ROI section, 2026](#)

Directional dashboard for cohort cost and PR volume comparison



Agentic Economics

THE AGENTIC ECONOMICS: MAKING THE BUSINESS CASE

20–65×

labor cost differential — agent compute vs senior engineer (BLS 2024)

5%

of AI pilots deliver material P&L improvement — the capture problem is infrastructure

3.6 days

payback on issue lifecycle automation — the highest-ROI entry point (modeled)

240%

modeled 3-year ROI — contingent on instrumentation investment coming first

What is the biggest infrastructure gap preventing the organization from capturing the \$2–5/hour opportunity today?